

Project Unicorn

Executive Summary | 2015 to 2018 | Federica Lucarelli & Sophia Sager

Which products make us money, which lose us money, and is discounting the reason?

The short answer is yes, but only partly. Some products lose money regardless of how they are priced. Others are profitable by nature but are being sold at a loss because of how aggressively they are discounted. The data draws a clear line between the two.

Across 9,994 order lines and 531 cities, Unicorn generated **\$286,347 in total profit** on **\$2.3 million in sales**, an overall margin of 12.5%. Technology runs at 17.4% margin, Office Supplies at 17.0%, and Furniture at just 2.5%. That split matters, and the evidence below explains why.

Part 1: SQL Evidence

Five queries from the Unicorn database establish the factual foundation: the scale of the business, where profit is generated, and which individual products are destroying margin.

EVIDENCE 1 | SQL QUERY 04

How many cities do we operate in?

```
SELECT COUNT(DISTINCT shipping_city) AS num_of_cities FROM orders;
```

Result: 531 unique cities

Unicorn has achieved wide geographic reach. With 531 cities served, the opportunity now is not further expansion but making the existing network more profitable.

EVIDENCE 2 | SQL QUERIES 02 AND 03

What was the most profitable city in 2015?

```
SELECT o.shipping_city FROM order_details od JOIN orders o ON od.order_id = o.order_id
WHERE EXTRACT(YEAR FROM o.order_date) = 2015
GROUP BY o.shipping_city ORDER BY SUM(od.order PROFITS) DESC LIMIT 1;
```

Result: New York City, \$14,753 total profit in 2015

New York City sets the benchmark for profitable urban logistics at Unicorn. High order density keeps shipping costs manageable and supports full-price selling.

EVIDENCE 3 | SQL QUERY 07

What was the largest single order by sales in 2015?

```
SELECT od.order_id, SUM(od.order_sales) AS total_sales FROM order_details od
JOIN orders o ON od.order_id = o.order_id
WHERE EXTRACT(YEAR FROM o.order_date) = 2015
```

```
GROUP BY od.order_id ORDER BY total_sales DESC LIMIT 1;
```

Result: Order CA-2015-145317, \$23,660 in sales

This single order shows that high-value transactions are achievable. Large orders in Technology carry the best revenue-to-margin ratio and are more efficient than chasing volume through discounting.

EVIDENCE 4 | CUSTOM SQL: BOTTOM 5 PRODUCTS BY TOTAL PROFIT

Which products are losing the most money?

```
SELECT p.product_name, SUM(od.order_profits) AS total_profit,
       SUM(od.order_profits)/SUM(od.order_quantity) AS profit_per_unit,
       AVG(od.order_discount) AS avg_discount
FROM order_details od JOIN product p ON od.product_id = p.product_id
GROUP BY p.product_id ORDER BY total_profit ASC LIMIT 5;
```

Result: Cubify CubeX 3D Printer Double Head: (\$8,880) total, (\$987) per unit, 53% avg discount

These are not occasional bad transactions. They represent a consistent pattern of products sold well below cost. The table below shows all five.

Product	Total loss	Per unit	Avg discount
Cubify CubeX 3D Printer (Double Head)	(\$8,880)	(\$987)	53%
Lexmark MX611dhe Monochrome Laser Printer	(\$4,590)	(\$255)	40%
Cubify CubeX 3D Printer (Triple Head)	(\$3,840)	(\$960)	50%
Chromcraft Bull-Nose Oval Conference Tables & Bases	(\$2,876)	(\$107)	28%
Bush Advantage Collection Racetrack Conference Table	(\$1,933)	(\$59)	35%

EVIDENCE 5 | CUSTOM SQL: TOP PRODUCT BY TOTAL PROFIT

Which product generates the most profit?

```
SELECT p.product_name, SUM(od.order_profits) AS total_profit
FROM order_details od JOIN product p ON od.product_id = p.product_id
GROUP BY p.product_name ORDER BY total_profit DESC LIMIT 1;
```

Result: Canon imageCLASS 2200 Advanced Copier, \$25,200 total profit, avg discount 12%

The top-performing product is a high-value Technology item sold at a low average discount. It generates more profit than the entire Furniture category on its own. This is the profile the business should look to replicate.

Part 2: Google Sheets Evidence

The spreadsheet analysis uses a pivot table and combo chart built in Google Sheets to show profitability patterns across categories and months, complementing the SQL work with visual and aggregated evidence.

EVIDENCE 6 | GOOGLE SHEETS: PIVOT TABLE BY CATEGORY AND SUB-CATEGORY

Which sub-categories are profitable and which are loss-making overall?

Result: Tables (\$17,733 loss, -8.6% margin), Bookcases (\$3,479 loss, -3.0% margin), Supplies (\$1,187 loss, -2.5% margin). Grand Total: \$286,347 profit on \$2,297,355 sales at 12.46% overall margin.

The pivot table groups all order lines by order month, category and sub-category, showing total sales, total profit and profit margin for each combination across 2015 to 2018. Three sub-categories produce a net loss across the entire period. Tables is the most significant at -8.6% margin. The Grand Total row confirms the overall business figures used throughout this report.

Sub-category	Category	Total profit	Margin
Copiers	Technology	+\$55,618	37.2%
Paper	Office Supplies	+\$34,053	43.4%
Accessories	Technology	+\$41,932	25.0%
Phones	Technology	+\$44,492	13.5%
Tables	Furniture	(\$17,733)	(8.6%)
Bookcases	Furniture	(\$3,479)	(3.0%)
Supplies	Office Supplies	(\$1,187)	(2.5%)

EVIDENCE 7 | CUSTOM SQL: AVERAGE PROFIT BY DISCOUNT LEVEL

At what point does a discount stop being a sales tool and start creating losses?

```
SELECT
  od.order_discount,
  COUNT(*) AS num_lines,
  AVG(od.order_profits) AS avg_profit
FROM order_details od
GROUP BY od.order_discount
ORDER BY od.order_discount;
```

Result: At 0% discount, avg profit is +\$66.78. At 20% it is +\$24.95. At 30% it flips to (\$39.43). Every discount level above 20% produces a loss.

Your query results show the average profit per order line at each discount level. The numbers differ slightly from the CSV analysis because the SQL database and the spreadsheet dataset are not identical, as noted in the project brief. The pattern is the same in both: profit is positive up to and including 20% discount, then turns negative above it. The table below uses your SQL results.

Discount level	Avg profit per line	Number of lines	Outcome
0%	+\$66.78	4,793	Profitable
10%	+\$110.78	101	Profitable
15%	+\$27.23	52	Profitable
20%	+\$24.95	3,658	Profitable
30%	(\$39.43)	227	Loss
40%	(\$111.90)	206	Loss
50%	(\$310.73)	66	Loss
60%	(\$43.07)	138	Loss
70%	(\$95.91)	418	Loss
80%	(\$101.82)	300	Loss

EVIDENCE 8 | CUSTOM SQL: FURNITURE PROFIT SPLIT BY DISCOUNT APPLIED

Is Furniture a broken category, or is the problem how it is priced?

```
SELECT
  product_subcategory,
  CASE WHEN order_discount = 0 THEN 'No discount'
        ELSE 'Discount applied' END AS pricing,
  ROUND(AVG(order_profits)::numeric, 2) AS avg_profit,
  ROUND(SUM(order_profits)::numeric, 0) AS total_profit,
  COUNT(*) AS num_lines
FROM order_details
JOIN product ON order_details.product_id = product.product_id
WHERE product_category = 'Furniture'
GROUP BY product_subcategory, pricing
ORDER BY product_subcategory, pricing;
```

Result: Tables with no discount: avg +\$184.35, total +\$13,273 across 72 lines. Tables with discount applied: avg (\$125.53), total (\$31,006) across 247 lines.

This is the clearest proof that the problem is pricing, not product. Tables sold at full price return \$184.35 profit per line on average. The same product with any discount applied loses \$125.53 per line on average. The total swing for Tables alone is \$44,279 driven entirely by discount policy. Bookcases and Furnishings show the same pattern: both are profitable at full price and loss-making when discounted.

Sub-category	Pricing	Avg profit	Total profit	Lines
Bookcases	No discount	+\$101.22	+\$6,073	60
Bookcases	Discount applied	(\$56.86)	(\$9,552)	168
Furnishings	No discount	+\$29.28	+\$16,690	570
Furnishings	Discount applied	(\$9.81)	(\$3,786)	386
Tables	No discount	+\$184.35	+\$13,273	72

Tables	Discount applied	(\$125.53)	(\$31,006)	247
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Part 3: Tableau Visual Analysis

The Tableau analysis is structured across two dashboards. The first establishes the problem — where Unicorn is losing money and at what scale. The second explains the cause and points to the action. Each chart was designed to communicate one distinct dimension of the same argument.

Dashboard 1: We Are Selling Our Way to a Loss

Dashboard 1 combines the treemap and a geographic profit map to show the scale and location of the problem before the cause is introduced.

E9 Not All Products Are Created Equal — Treemap

Which sub-categories generate the most profit, and which are destroying it?

[Insert treemap screenshot here]

Size = Sales volume · Colour = Profit (dark green = high profit, red = loss)

Why it matters: The treemap encodes two variables simultaneously. Rectangle size represents total sales volume and colour represents profit, from dark green to red. Tables is immediately striking: large rectangle meaning significant sales, but deep red meaning loss-making at scale. Copiers, Accessories and Phones dominate the green. The visual makes the core argument in a single glance — Tables is not marginal. It is a material drag on profitability.

E10 The Bigger the Discount, the Greater the Loss — Highlight Table

At which discount band does each sub-category tip into loss, and by how much?

The Higher the Discount, the Deeper the Loss
Profit by Sub-Category and Discount Level



Profit by sub-category and discount band · Green = profitable · Red = loss

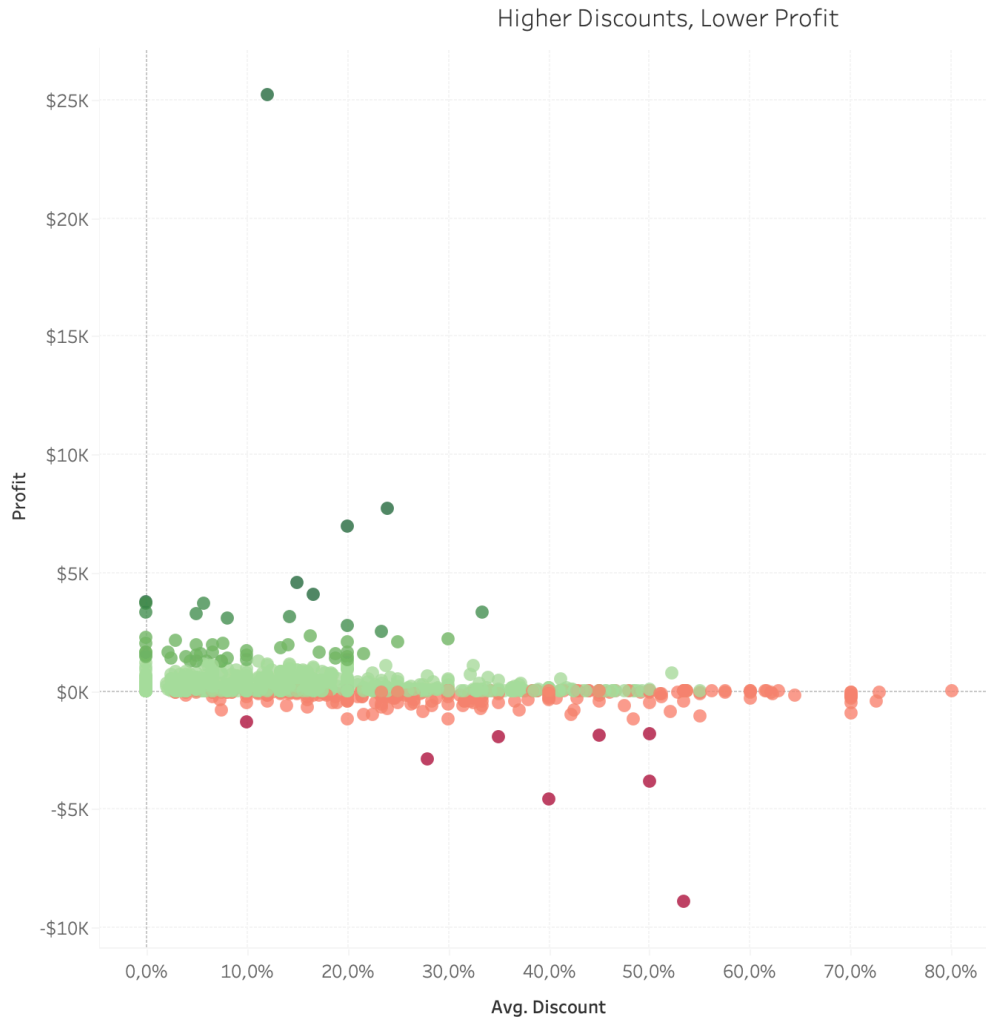
Why it matters: Sub-categories are grouped into three discount bands: Low (0 to 15%), Mid (15 to 30%) and High (30%+). Tables appear only in the Mid bucket and are already deeply red — it loses money at even moderate discounts. Most profitable sub-categories generate all their returns in the Low bucket. This visually confirms the 20% threshold from the SQL analysis and shows that the solution does not require eliminating discounts, just capping them.

Dashboard 2: The Cause and the Action

Dashboard 2 presents the scatter plot and the inventory bar chart together. The scatter proves discounting is the mechanism. The bar chart names the specific products to act on.

E11 Do Discounts Drive Profit or Sales Loss? — Scatter Plot

Is there a consistent relationship between discount level and profit across all orders?

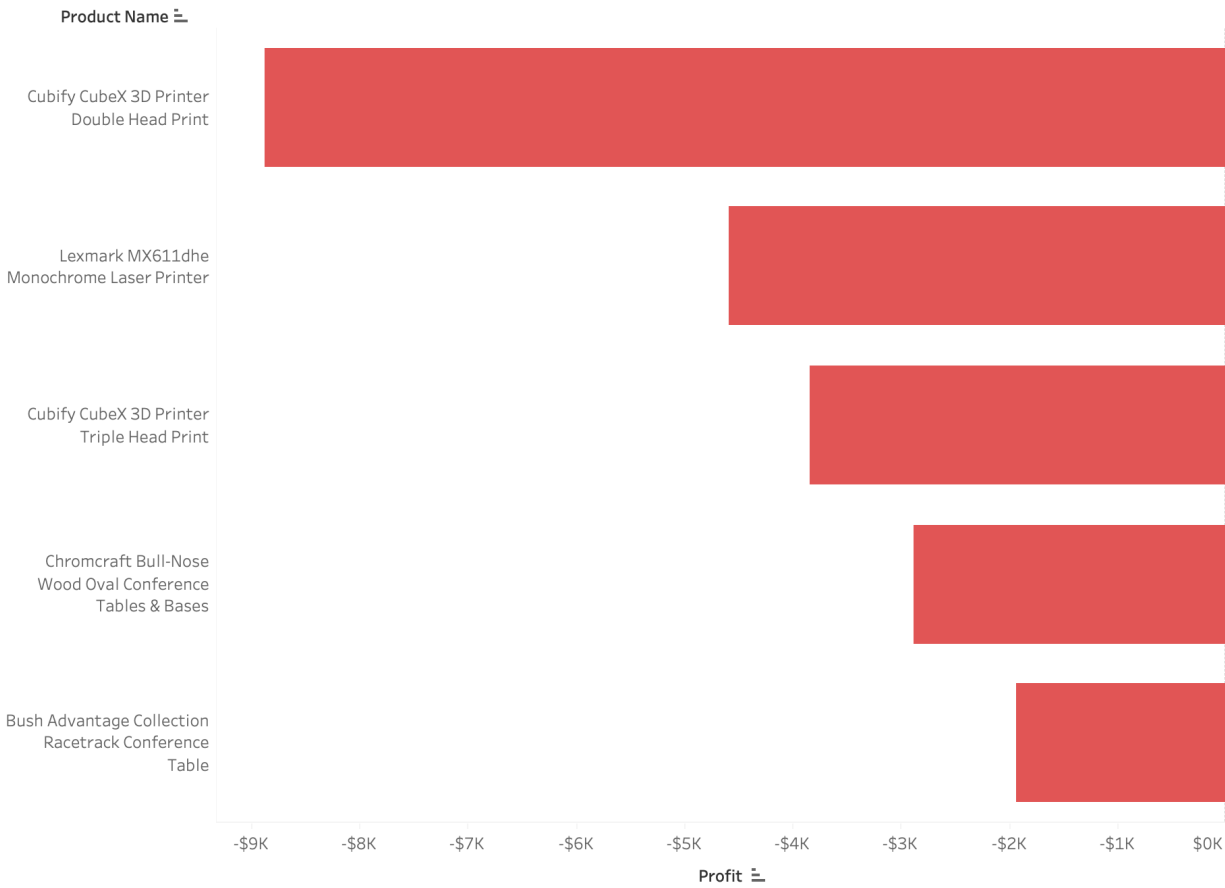


Why it matters: Dots cluster in positive profit territory at low discounts and shift toward negative as discount increases. The chart also allows toggling between profit and sales on the y-axis, revealing that sales volume does not collapse at full price. This directly undermines the common argument that deep discounts are necessary to move product. The data shows they are not driving volume, only destroying margin.

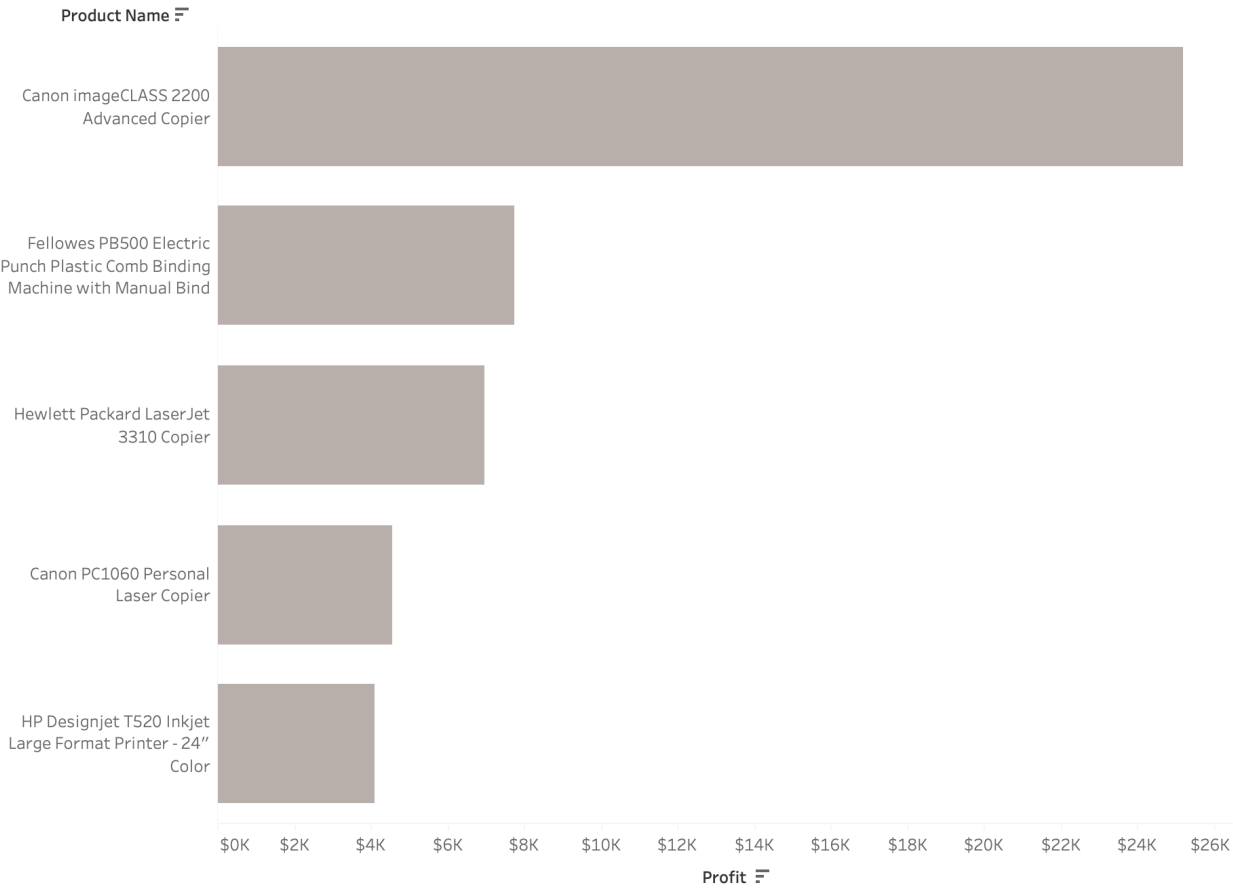
E12 Inventory Optimization: Identifying High-Loss Products — Bar Chart

Which specific products are driving the losses, and how large is each one?

Profit & Loss by Product *(toggle between top and bottom products)*



Profit & Loss by Product (toggle between top and bottom products)



Profit by product · Grey = profitable · Red = loss · Filterable by category and sub-category

Why it matters: The chart names the specific products to act on. Filtered to Furniture, Tables products dominate the red bars with the Chromcraft Bull-Nose Oval Conference Tables producing the largest single-product loss. Filtered to Technology, the Cubify CubeX printers and the Lexmark MX611dhe stand out as loss-makers regardless of category. The bar chart converts the analysis into a concrete list of decisions, making it the natural conclusion to the dashboard.

Key Insights

- Technology and Office Supplies are the engine of the business, running at 17.4% and 17.0% margin respectively and together accounting for \$267,903 of total profit.
- Furniture is not a broken category. It is a mispriced one. At full price it averages \$69.53 profit per order line. With any discount applied, that flips to a (\$30.88) average loss.
- The 20% discount level is the single most important threshold in the data. Every tier below it produces positive average profit. Every tier above it produces a loss, worsening as the discount increases.

- 1,393 order lines currently sit above the 20% discount threshold, each generating a loss on average. This is the most direct and addressable source of profit leakage in the business.
- A small number of products are genuinely loss-making at any price. The Cubify CubeX 3D Printer (Double Head) has lost \$8,880 in total at (\$987) per unit on a 53% average discount.
- The highest-profit product in the portfolio, the Canon imageCLASS 2200 Advanced Copier at \$25,200, is sold at an average discount of just 12%. It outperforms the entire Furniture category on its own.
- Q4 volume growth is being bought with margin compression. The monthly trend shows profit margin declining during the highest-revenue quarter, suggesting the holiday strategy relies on discounting to hit revenue targets.

Conclusion and Recommendations

Unicorn is a profitable business being held back by a specific and fixable pricing behaviour. Technology and Office Supplies generate strong, consistent returns. Furniture, which the data shows is profitable at full price, is being undermined by discounting that turns positive-margin products into loss-makers at scale. A small cluster of products lose money at any price and should be evaluated for removal from the catalogue.

The clearest implication is that growing revenue through deeper discounts is counterproductive. The 1,393 lines above the 20% threshold are not driving profitable growth. Three actions would have the largest immediate impact. First, setting a 20% discount ceiling with sign-off required above it directly addresses the largest source of loss. Second, reviewing the five loss-making products for discontinuation removes items that lose money regardless of pricing. Third, introducing a minimum margin requirement on Furniture items before a discount can be approved would recover the majority of the \$44,000 swing identified in the Tables analysis, without requiring any change to the product catalogue itself.